3.1 Techniques to test/check and review digital games	
Techniques to test/check the technical properties of digital games ◻ Methods of testing and checking ▪ test plan ▪ types of test data ○ normal ○ extreme ○ exceptional ◻ Elements of digital games to test/check ▪ test input or behaviours ○ trying to cheat or break ○ testing by playing to the rules or by deliberately breaking the rules ▪ functionality tests ○ navigation features ○ interactive features ○ inputs and outputs ○ scoring, timing, game start and game end ○ object movement ○ geometric parameters	To include: • The structure, content and use of test plans, checklists and success criteria • How to record test results and how and when to retest • How and why to test iteratively both during production and post-production • Planning and carrying out a range of functionality tests to make sure digital games function as intended • The importance of testing each variable as it is amended or introduced and recording the results of tests/keeping test logs • Carrying out retests and using version control to rollback features to earlier states where necessary • Determining acceptable glitches or errors and how to prioritise fixing errors or correcting programming where tests have failed Does not include: • Peer or client testing/focus group feedback • Performance tuning and debugging, black and white box testing, soak testing, load testing, regression testing
Techniques to review the fitness for purpose of a Game Design Documents (GDDs) ◻ Suitability for client requirements ◻ Suitability for target audience ▪ suitability and detail of content ▪ clarity of explanation ◻ Review of originality, appeal, engagement and audio-visual quality	To include: • Strengths and weaknesses of created digital games • Comparing created digital games against concepts outlined in GDDs • Assessing the appropriateness of chosen style and approaches/conventions of GDDs and digital games for clients and target audiences • Assessing fitness for purpose of GDDs and digital games in terms of audience engagement and functionality e.g. digital games should be playable, GDDs should outline digital game concepts Does not include: • Client/peer/focus group feedback or review, questionnaires, vox pop

Unit R099: Digital games	
3.2 Improvements and further developments	
Constraints which limit the effectiveness of digital games □ Digital game constraints ▪ time ▪ resource ▪ hardware ▪ software ▪ skills □ Digital game improvements ▪ fixing errors to improve gameplay, mechanics and geometric parameters ▪ improving visual and/or audio components to increase player engagement ▪ altering gameplay difficulty or scoring mechanisms to improve player experience	To include: • How the quality of digital games is constrained by time, resources, hardware, software, budget, legislation, skills • The feasible improvements to digital games in terms of client requirements and target audience engagement • How to identify which changes to the programming, visuals and audio of digital games would make the most difference to player engagement and experience
Further development opportunities for digital games □ Further developments ▪ appearance and style ▪ gameplay ▪ narrative or objectives ▪ interactions ▪ player feedback/outputs ▪ difficulty ▪ downloadable content ▪ expansion packs ▪ second game instalment	To include: • How successful digital games can lead to repeat business/further commissions from a client • How different resources, software, budget and skills could help digital games to be developed further • How to devise further developments in terms of client requirements and target audience • How to ascertain which elements of an MVP can be developed and the order of priority for such developments

Marking criteria

Section 6.4 provides full information on how to mark the NEA units and apply the marking criteria. The marking criteria descriptor words are further explained in Appendix B.

The tables below contain the marking criteria for the tasks for this unit. If a student's work does not meet any Mark Band 1 (MB1) criteria for the task, you must award zero marks for that task.

Unit R099 – Topic Area 1: Plan digital games		
MB1: 1–2 marks	**MB2: 3–4 marks**	**MB3: 5–6 marks**
Produces a **basic** interpretation of the client brief. Explanation of how the intended product meets the client brief and why it appeals to the target audience is **limited**.	Produces an **adequate** interpretation of the client brief. Explanation of how the intended product meets the client brief and why it appeals to the target audience is **sound**.	Produces an **effective** interpretation of the client brief. Explanation of how the intended product meets the client brief and why it appeals to the target audience is **comprehensive**.
MB1: 1–3 marks	**MB2: 4–6 marks**	**MB3: 7–8 marks**
Produces **basic** pre-production and planning documentation. Pre-production and planning documentation support the creation of **few** elements of the final product.	Produces **adequate** pre-production and planning documentation. Pre-production and planning documentation support the creation of **some** elements of the final product.	Produces **detailed** pre-production and planning documentation. Pre-production and planning documentation support the creation of **all** elements of the final product.
MB1: 1–2 marks	**MB2: 3–4 marks**	**MB3: 5–6 marks**
Demonstrates **limited** understanding of how assets will contribute to the effectiveness of the final product.	Demonstrates **sound** understanding of how assets will contribute to the effectiveness of the final product.	Demonstrates **comprehensive** understanding of how assets will contribute to the effectiveness of the final product.

MB1: 1–4 marks	MB2: 5–8 marks	MB3: 9–12 marks
Use of technical skills to create the component parts is **limited** in its effectiveness. Conventions and creativity in the components are **limited** in their fitness for purpose. The range of components supports the creation of the final product in a **limited** way.	Use of technical skills to create the component parts is **partly** effective. Conventions and creativity in the components are **adequate** in their fitness for purpose. The range of components **partly** supports the creation of the final product.	Use of technical skills to create the component parts is **effective**. Conventions and creativity in the components are **fully** fit for purpose. The range of components **fully** supports the creation of the final product.
MB1: 1–5 marks	**MB2: 6–10 marks**	**MB3: 11–14 marks**
Use of technical skills to create the final product is **limited** in its effectiveness. Conventions and creativity are applied in the final product in a **limited** way. Final product is **limited** in its fitness for purpose.	Use of technical skills to create the final product is **partly** effective. Conventions and creativity are **adequately** applied in the final product. Final product is **adequately** fit for purpose.	Use of technical skills to create the final product is **effective**. Conventions and creativity are **effectively** applied in the final product. Final product is **fully** fit for purpose.
MB1: 1–3 marks	**MB2: 4–6 marks**	**MB3: 7–8 marks**
Formats of the saved/exported components are **limited** in their appropriateness. Properties and format(s) of the final product are **limited** in their appropriateness.	Formats of the saved/exported components are **adequate** in their appropriateness. Properties and format(s) of the final product are **adequate** in their appropriateness.	Formats of the saved/exported components are **clearly** appropriate. Properties and format(s) of the final product are **clearly** appropriate.

MB1: 1–3 marks	MB2: 4–7 marks	MB3: 8–10 marks
Testing/checking is **limited** in its effectiveness in reviewing technical properties. Review demonstrates **limited** understanding of the effectiveness of the final product for client and target audience.	Testing/checking is **partly** effective in reviewing technical properties. Review demonstrates **sound** understanding of the effectiveness of the final product for client and target audience.	Testing/checking is **fully** effective in reviewing technical properties. Review demonstrates **critical** understanding of the effectiveness of the final product for client and target audience.
MB1: 1–2 marks	**MB2: 3–4 marks**	**MB3: 5–6 marks**
Recommendations demonstrate **limited** understanding of areas for improvement and further development. Recommendations have **limited** explanation.	Recommendations demonstrate **sound** understanding of areas for improvement and further development. Recommendations are **partly** explained.	Recommendations demonstrate **comprehensive** understanding of areas for improvement and further development. Recommendations are **fully** explained.

Assessment guidance

Task	Assessment guidance
Task 1	**Strand 1a** For Mark Band (MB)1, students may have simply stated the chosen audience and restated the client. An explanation of a single game idea would fit into MB1, whereas if several ideas are outlined and the preferred option chosen this would be more appropriate for MB2 or MB3. An explanation of only one or two ways in which the game idea meets the client brief and appeals to the target audience would be suitable for MB1. MB3 could be achieved by explaining a few ways in great detail (depth) or by explaining many ways concisely (breadth). Diagrammatic representations would illustrate the interpretation but would not be sufficient on their own to fulfil the requirement to explain for the upper mark bands. When interpreting the brief, students need to make decisions independently. Although it is to be expected that different students may make similar decisions and develop similar ideas, it would be highly unusual for all students in a cohort to have identical work. **Strand 1b** Pre-production documentation for the game idea could be presented in a number of different formats. Basic planning for MB1 may cover only the main points of the game, whereas detailed planning for MB3 would be expected to include depth. It would be unlikely that any single pre-production document would cover 'all' elements of the final game, and different types of planning document would be expected in order to meet the criteria for the upper mark bands. Students must not be directed to complete specific planning tasks but may be referred to the teaching and learning content for the unit. When planning their digital game and justifying their design choices, students need to make decisions independently. Although it is to be expected that different students may make similar decisions and develop similar ideas, it would be highly unusual for all students in a cohort to have identical work.

	Strand 1c This strand requires students to demonstrate understanding of how assets contribute to the effectiveness of the game. A list of assets would therefore be suitable for MB1, but would not demonstrate understanding. A consideration of the technical properties of the assets chosen and reference to how the style or type of assets would appeal to the target audience or match the genre of the game would contribute to meeting the requirements for MB3.
Task 2	**Strand 2a** The Game Design Document (GDD) may briefly outline a few aspects of the game for MB1, but at MB2 and above should include explanations and could include diagrams and illustrations to improve clarity. The conventions of Game Design Documents would be expected to be applied to meet the requirements of MB3 in terms of the structure, format and content of the GDD. To be fit for purpose the GDD should both inform the client and promote the game idea and concept. Assets as components for the game support the creation of the product by virtue of their range and their properties. To fully support the creation of the game for MB3, assets would need to be complete and technically fit for purpose. Few assets, or many assets but with limited technical suitability would be appropriately assessed in MB1. **Strand 2b** This strand relates to the students' playable game. For MB1, the game may be simplistic, incomplete or limited in its functionality. Some evidence of the tools used to create the game would support marks for this strand, even if the game is not finished or exported. Students can be given credit for presenting screenshots to show the tools of the game creation software they used, especially as some software may not be available to assessors. A game which functions correctly but is not original would fit the criteria for MB1 in this strand for the creativity and technical skills used. For MB3, the game should be playable in order to meet the criteria for fitness for purpose. Evidence to support this could comprise video or screen recording of the game being played by the student, especially if the skill level required to play the game is high. When creating their digital game, students must make decisions independently. They must apply what they have learnt and not be led through a process to create a digital game. Students must not be directed to use particular software or software tools and/or techniques. **Strand 2c** This strand assesses the appropriateness of the components (the GDD and the assets) and the final product (the finished game). In order to meet the criteria for MB3, the GDD, assets and the playable game should all be saved using appropriate formats and properties. The GDD could be presented as a collection of separate files at MB1, whereas for MB3 the different elements could be collated into a single document or product which is named and saved suitably for the client to access. Where the final game is not exported from the native software, MB1 would be appropriate. For MB3, there should be clear evidence of saving/exporting in a format which is fully suitable for all the aspects of the client brief and the assessment requirements. When exporting their digital game, students need to independently decide on suitable electronic formats and properties. Students can provide evidence of this by either submitting the exported final product or a video or screen recording clearly showing how they exported their final product.

Task 3	Students must produce their own review. They must apply what they have learnt and not be led through a process of reviewing their completed digital game and concept. **Strand 3a** Evaluation of the technical properties of the finished game could be presented as a completed test plan. At MB1 this may only outline a few tests, or may be limited by testing only some aspects. If results and retests are considered this would contribute to MB2, whereas to be fully effective for MB3 the technical properties should be considered comprehensively. An outline of how errors were resolved or prioritised for resolution might be expected at the upper end of the mark range. The second descriptor assesses understanding of the effectiveness of the game and GDD for the client and target audience. If only the client or target audience are considered, a mark in MB1 may be suitable, whereas for MB3 both client and target audience, and the game and its concept should be considered. A test plan on its own would not evidence understanding of effectiveness, only functionality; so, some form of account or explanation would be expected for MB2 and MB3. **Strand 3b** This strand requires an understanding of areas for improvement and further development, and an explanation of the recommendations. A list of aspects which failed when testing the game would demonstrate limited understanding for MB1. For MB3 students may have prioritised aspects to be addressed. Explaining which are the most important and why might be one way to demonstrate comprehensive understanding. Areas for improvement and further development should be explained with reference to the client requirements and target audience engagement in order to demonstrate comprehensive understanding for MB3, and should include both the playable game and the concept as set out in the GDD. If the game or GDD only are considered, a mark in MB1 would be suitable. Since there is no requirement for a prescribed number of improvements and developments, a few aspects considered and explained in detail could nevertheless meet the criteria for MB3.

Synoptic assessment

Some of the knowledge, understanding and skills required when completing this unit will draw on the learning developed in Unit R093. The following table details where these synoptic links can be found:

R099: Digital games		R093: Creative iMedia in the media industry	
Topic Area		**Topic Area**	
1	Plan digital games	2	Factors influencing product design
		3	Pre-production planning
2	Create digital games	3	Pre-production planning
		4	Distribution considerations
3	Review digital games	2	Factors influencing product design
		3	Pre-production planning
		4	Distribution considerations

More information about synoptic assessment within this qualification can be found in section 5.2 Synoptic assessment.

J834	OCR Level 1/Level 2 Cambridge National in Creative iMedia	120*	603/7090/7
Made up of three mandatory units: • Units R093 and R094 • And one from R095, R096, R097, R098 or R099			

*the GLH includes assessment time for each unit

Unit R093: Creative iMedia in the media industry	
48 GLH 1 hour 30 minute written examination 70 marks (80 UMS) OCR set and marked Calculators are not required in this exam	This question paper has two parts: • Part A – includes closed response, multiple choice and short answer questions which assess PO1 [10 marks]. • Part B – includes closed response, short answer and three extended response questions. Two of the extended response questions will assess PO3 and the third, PO1/PO2. All part B questions relate to a single scenario [60 marks].
Unit R094: Visual identity and digital graphics	
30 GLH OCR-set assignment 50 marks (50 UMS) Centre-assessed and OCR moderated	This set assignment contains two practical tasks. It should take approximately 10-12 GLH to complete.
Unit R095: Characters and comics	
42 GLH OCR-set assignment 70 marks (70 UMS) Centre-assessed and OCR moderated	This set assignment contains three practical tasks. It should take approximately 12-15 GLH to complete.
Unit R096: Animation with audio	
42 GLH OCR-set assignment 70 marks (70 UMS) Centre-assessed and OCR moderated	This set assignment contains three practical tasks. It should take approximately 12-15 GLH to complete.
Unit R097: Interactive digital media	
42 GLH OCR-set assignment 70 marks (70 UMS) Centre-assessed and OCR moderated	This set assignment contains three practical tasks. It should take approximately 12-15 GLH to complete.

42 GLH OCR-set assignment 70 marks (70 UMS) Centre-assessed and OCR moderated	This set assignment contains three practical tasks. It should take approximately 12-15 GLH to complete.

The terminal assessment rule or 'terminal rule' means that the exam must be taken in the final assessment series of the student's course. Non examined assessment (NEA) units can be submitted in the same series as the exam or an earlier series but the exam must be taken in the final series.

If a student takes the exam in a series before their work for NEA units are submitted, this is considered as a 'practice' attempt and will not contribute to their final qualification grade. The student must take their exam again in their final assessment series and that exam result will be used towards the student's final qualification grade.

For more information on the terminal rule and also what it means for performance tables see Section 7.3. Resitting units before certification and Retaking the qualification.
OCR-set assignments for units R094-R099 are available on our secure website for teachers, Teach Cambridge.

5.2 Synoptic assessment

Synoptic assessment is a built-in feature of this qualification. It means that students need to use an appropriate selection of their knowledge, understanding and skills developed across the qualification in an integrated way and apply them to a key task or tasks.

This also helps students to build a holistic understanding of the subject and the connections between different elements of learning, so they can go on to apply what they learn from this qualification to new and different situations and contexts.

The externally assessed unit R093 allows students to gain underpinning knowledge and understanding relevant to the digital media industry, and the non examined assessment (NEA) units R094 - R099 draw on and strengthen this learning by letting students apply their learning in a practical, skills-based way.

It is important to be aware of the synoptic links between the units so that teaching, learning and assessment can be planned accordingly. Then students can apply their learning in ways which show they are able to make connections across the qualification when they are assessed.

5.3 Transferable skills

This qualification also allows students the opportunity to gain broad, transferable skills and experiences that can be applied as they progress into their next stages of study and life and to enhance their preparation for future employment.

Students will have the opportunity to develop the following skills that are transferable to different real-life contexts, roles or employment:

- Problem Solving (PS) - within the NEA units students will learn about the tools and techniques used to create digital media products. This will include techniques to record ideas, plan solutions and review outcomes to check if the requirements of clients and audiences/consumers are met.
- Analytical Skills (AS) – students will learn how to analyse scenarios and work out who clients, audiences/consumers are and what they require from digital media products. They will also learn written analysis skills through the review of pre-production documents and the digital media products created.

- Digital Presentation (DP) – throughout the Creative iMedia qualification students will learn to identify and make use of the tools and techniques appropriate to the digital media product they are planning and creating. This will include the selection of appropriate media (text, images, audio or video) to convey meaning, create impact and/or engage audiences.
- Planning (P) – students will learn planning techniques within the EA unit relevant to the media industry but also applicable much more generally.
- Creative Thinking (CT) - within each NEA unit students will learn about the different forms of creativity and how creativity is integral to producing effective digital media products. This will involve them exploring and generating ideas, making connections to find imaginative solutions and outcomes that add value.

5.4 Grading and awarding grades

All results are awarded on the following scale:

- Distinction* at Level 2 (*2)
- Distinction at Level 2 (D2)
- Merit at Level 2 (M2)
- Pass at Level 2 (P2)
- Distinction at Level 1 (D1)
- Merit at Level 1 (M1)
- Pass at Level 1 (P1).

The shortened format of the grade will show within results files and results reports. However, the full format of the grade will be on the certificates issued to students.

The boundaries for Distinction at Level 2, Pass at Level 2, and Pass at Level 1 are set judgementally. Other grade boundaries are set arithmetically.

The Merit (Level 2) is set at half the distance between the Pass (Level 2) grade and the Distinction (Level 2) grade. Where the gap does not divide equally, the Merit (Level 2) boundary is set at the lower mark (For example, 45.5 would be rounded down to 45).

For the examined unit, the Distinction* (Level 2) grade is normally set at about 0.75 of the D2-M2 distance above the D2 boundary mark.

To set the Distinction (Level 1) and Merit (Level 1) boundaries, the gap between the Pass (Level 1) grade and the Pass (Level 2) grade is divided by 3, and the boundaries set equidistantly. Where this division leaves a remainder of 1, this extra mark will be added to the Distinction (Level 1) to Pass (Level 2) interval, meaning the Distinction (Level 1) boundary will be lowered by 1 mark. Where this division leaves a remainder of 2, the extra marks will be added to the Distinction (Level 1) to Pass (Level 2) interval, and the Merit (Level 1) to Distinction (Level 1) interval, meaning the Distinction (Level 1) boundary will be lowered by 1 mark, and the Merit (Level 1) boundary will be lowered by 1 mark.

For example, if Pass (Level 2) is set judgementally at 59, and Pass (Level 1) is set judgementally at 30, then Distinction (Level 1) is set at 49, and Merit (Level 1) is set at 39.

Grades are indicated on qualification certificates. However, results for students who fail to achieve the minimum grade (Pass at Level 1) will be recorded as unclassified (U or u) and **will not** be shown on certificates.

This qualification is unitised. Students can take units across different series and can resit units (see section 7.8 Unit and qualification resits). Grade boundaries are set per unit, per series, so may be set in different places for a unit in different series. When working out students' overall grades, OCR needs to be able to compare performance on the same unit in different series when different grade boundaries may have been set, and between different units. We use a Uniform Mark Scale (UMS) so this can be done.

A student's uniform mark for each unit is calculated from the student's raw mark on that unit. The raw mark boundary marks are converted to the equivalent uniform mark boundary. Marks between grade boundaries are converted on a pro rata basis.

When unit results are issued, the student's unit grade and uniform mark are given. The uniform mark is shown out of the maximum uniform mark for the unit (for example, 42/60).

The table below shows the Raw marks and UMS marks for each unit:

	Exam	NEA1	NEA2

30	50	45	40	35	30	25	20	15	**0**
42	70	63	56	49	42	35	28	21	**0**
48	80	72	64	56	48	40	32	24	**0**

The student's uniform mark for Unit R093 will be combined with the uniform mark for the NEA units to give a total uniform mark for the qualification.

The student's overall grade will be determined by the total uniform mark. The following table shows the minimum total mark for each overall grade:

Max Uniform Mark	Qualification Grade							
	Distinction* at L2	Distinction at L2	Merit at L2	Pass at L2	Distinction at L1	Merit at L1	Pass at L1	**U**
200	180	160	140	120	100	80	60	**0**

On results day we will publish a marks calculator, for the specific series, on our website to help you convert raw marks/number of achieved criteria into uniform marks.

5.5 Performance descriptors

Performance descriptors give a general indication of likely levels of attainment by representative students performing at boundaries: Distinction at Level 2, Pass at Level 2 and Pass at Level 1.

Performance descriptor – Distinction at Level 2

Students will be able to:

- recall a wide range of information relating to the media industry and the planning, creating and reviewing of digital media products
- perceptively evaluate the purpose and uses of digital media products
- understand and use a wide range of media related terminology correctly
- demonstrate analytical and evaluative skills
- interpret and present information with sensitivity to needs and with a flair for effective communication
- work independently and manage time efficiently
- use techniques efficiently to source, select and store appropriate assets effectively, in a wide variety of contexts
- create solutions which demonstrate detailed consideration of target audience and for a specific brief
- confidently use and apply a wide range of techniques to create work that is fit for purpose
- perceptively analyse problems encountered in a media context.

Performance descriptor – Pass at Level 2

Students will be able to:

- recall a range of information relating to the media industry and the planning, creating and reviewing of digital media products
- evaluate the purpose and uses of digital media products
- understand and use a range of media related terminology correctly
- demonstrate analytical and evaluative skills
- present information with awareness of needs and communication
- work independently and manage time efficiently
- create solutions which demonstrate consideration of target audience and for a specific brief
- use techniques to source, select and store appropriate assets, in a variety of contexts
- use and apply a range of techniques to create work that is fit for purpose
- analyse problems encountered in a media context.

Performance descriptor – Pass at Level 1

Students will be able to:

- recall some information relating to the media industry and the planning, creating and reviewing of digital media products
- understand the purposes and uses of digital media products
- understand and use some media related terminology correctly
- demonstrate some evidence of basic analytical and evaluative skills
- present information with some awareness of needs
- work with guidance to given timescales
- create solutions which demonstrate some awareness of target audience and a specific brief
- use techniques to source, select and store information
- use and apply some techniques to create work that is suitable for a specific brief
- demonstrate an understanding of some problems encountered in a media context.

6 Non examined assessment (NEA) units (R094 – R099)

This section provides guidance on the completion of the NEA units (R094 – R099). The NEA units are designed so that students can build a portfolio of evidence to meet the topic areas for the unit.

Assessment for this qualification must adhere to JCQ's Instructions for conducting non-examination assessments (Vocational and Technical Qualifications).

Please **do not** use JCQ's Instructions for Conducting Non-examination Assessments (GCE and GCSE Specifications) – these are only relevant to A Level and GCSE specifications.

Units R094 – R099 are centre assessed and externally moderated by us.

You **must** make sure that you have read and understood all of the rules and guidance provided in this section **before** your students complete and you assess the set assignments.

If you have any queries please contact us for help and support.

6.1 Preparing for NEA unit delivery and assessment